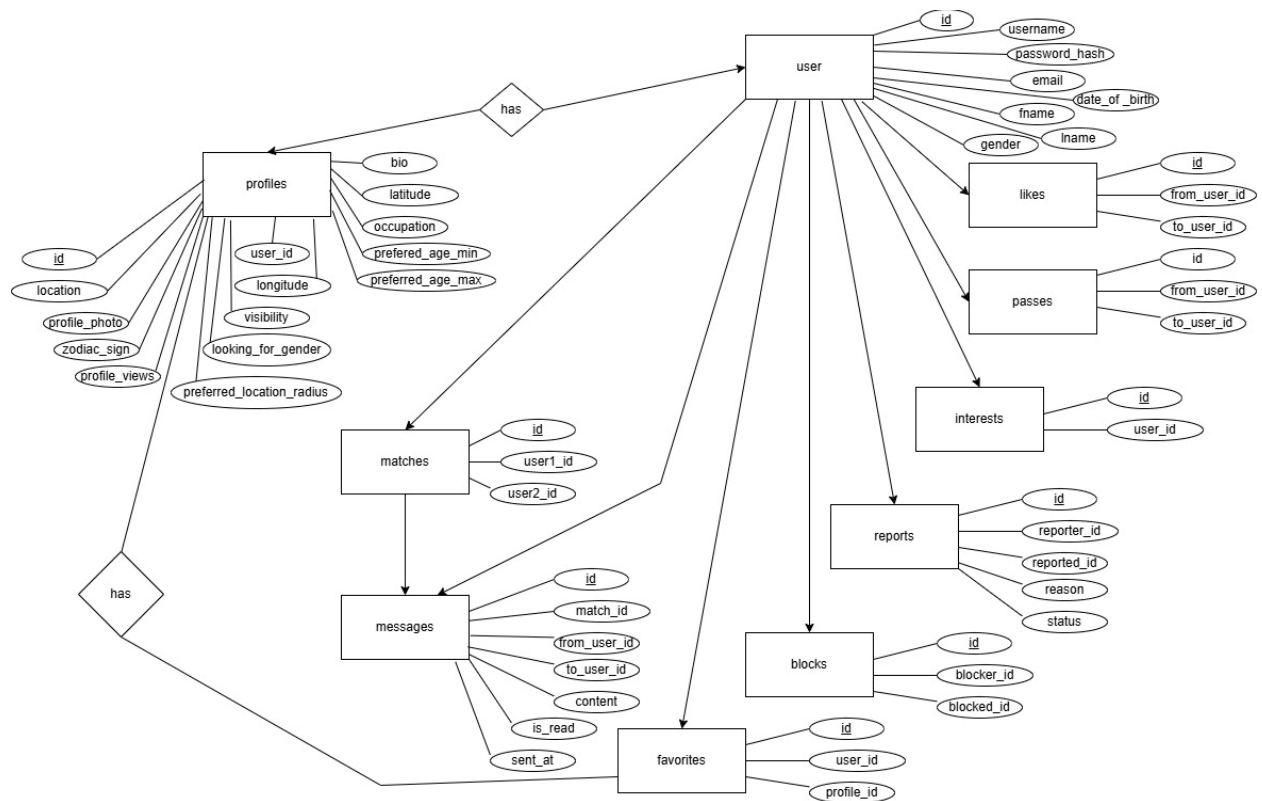


ERD DIAGRAM AND **DATABASE DOCUMENTATION**

ER Diagram



Database Schema Documentation

Database: Relational (SQLAlchemy / Flask) Total Tables: 10

Overview

This database supports the dating application. It handles user authentication, profile management, matchmaking (likes, passes, matches), messaging between matched users, safety features (reports, blocks) and where to find images uploaded by users.

Tables

1. users

The main table. Every other table references this one.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
username	String(80)	Unique, Not Null
email	String(120)	Unique, Not Null
password_hash	String(128)	Not Null
fname	String(60)	Not Null
lname	String(60)	Not Null
gender	String(1)	Not Null
date_of_birth	Date	Not Null
created_at	DateTime	Default: current timestamp
updated_at	DateTime	Default: current timestamp, updates on change

Notes: Age is a calculated from date_of_birth.

2. profiles

Stores extended profile information for each user.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
user_id	Integer	Foreign Key → users.id, Unique, Not Null
bio	Text	Nullable
location	String(200)	Nullable
latitude	Float	Nullable
longitude	Float	Nullable
profile_photo	String(300)	Nullable
visibility	Boolean	Default: True
occupation	String(100)	Nullable
zodiac_sign	String(20)	Nullable
preferred_location_radius	Integer	Default: 50
preferred_age_min	Integer	Default: 18
preferred_age_max	Integer	Default: 99
looking_for_gender	String(20)	Default: 'all'
profile_views	Integer	Default: 0
created_at	DateTime	Default: current timestamp
updated_at	DateTime	Default: current timestamp, updates on change

3. interests

Stores individual interests/hobbies tagged to a user. A user can have many interests.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
user_id	Integer	Foreign Key → users.id, Not Null
interest	String(80)	Not Null
created_at	DateTime	Default: current timestamp

4. likes

Records when one user likes another user's profile.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
from_user_id	Integer	Foreign Key → users.id, Not Null
to_user_id	Integer	Foreign Key → users.id, Not Null
created_at	DateTime	Default: current timestamp

5. matches

Created when two users have mutually liked each other. Serves as the basis for messaging.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
user1_id	Integer	Foreign Key → users.id, Not Null
user2_id	Integer	Foreign Key → users.id, Not Null
matched_at	DateTime	Default: current timestamp

Unique Constraint: When removing a match the associated messages are deleted

6. messages

Stores messages exchanged between two matched users.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
match_id	Integer	Foreign Key → matches.id, Not Null
from_user_id	Integer	Foreign Key → users.id, Not Null
to_user_id	Integer	Foreign Key → users.id, Not Null
content	Text	Not Null
is_read	Boolean	Default: False
sent_at	DateTime	Default: current timestamp

Notes: Messages are tied to a match, meaning only matched users can message each other. And unmatching removes messages.

7. reports

Allows users to report other users for inappropriate behavior.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
reporter_id	Integer	Foreign Key → users.id, Not Null
reported_id	Integer	Foreign Key → users.id, Not Null
reason	Text	Not Null
status	String(20)	Default: 'pending'
created_at	DateTime	Default: current timestamp

8. blocks

Allows users to block other users, preventing interaction.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
blocker_id	Integer	Foreign Key → users.id, Not Null
blocked_id	Integer	Foreign Key → users.id, Not Null
created_at	DateTime	Default: current timestamp

9. favorites

Allows users to save profiles they are interested in.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
user_id	Integer	Foreign Key → users.id, Not Null
profile_id	Integer	Foreign Key → profiles.id, Not Null
created_at	DateTime	Default: current timestamp

10. passes

Records when a user passes another user's profile.

Column	Type	Constraints
id	Integer	Primary Key, Auto-increment
from_user_id	Integer	Foreign Key → users.id, Not Null
to_user_id	Integer	Foreign Key → users.id, Not Null
created_at	DateTime	Default: current timestamp